

NADINE MLAYEH

Software Engineer - Full-Stack (Java - Spring Boot | React - Angular)

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PROFILE

Software engineer (ISIMM, 2026) who builds and ships full-stack applications end to end - Angular and React front-ends, Spring Boot and NestJS APIs, and the data models underneath. Most recently built four microservices for an enterprise after-sales platform deployed on Kubernetes, with centralized Keycloak authentication. Also build applied AI features with the Gemini API, including a retrieval-augmented Q&A system over legal documents and structured job-posting extraction. Looking for a junior full-stack or backend role on a team that ships regularly.

TECHNICAL SKILLS

Backend: Java, Spring Boot, NestJS, REST API design, JPA/Hibernate, Prisma, JDBC

Frontend: React, Next.js, TypeScript, Angular, JavaScript, Tailwind CSS, Three.js / R3F

Databases: MySQL, PostgreSQL, MongoDB, Supabase

AI Integration: Google Gemini API, retrieval-augmented generation (RAG), embeddings, prompt design, structured JSON output

Architecture: Microservices, RabbitMQ (messaging), Keycloak (OAuth2/OIDC), MinIO (object storage)

Tools & Delivery: Git/GitHub, Docker, GitLab CI, Vercel, Render

Familiar with: Kubernetes, ArgoCD, AWS, Python, C/C++, Flutter/Dart

PROFESSIONAL EXPERIENCE

Software Engineering Intern - Final-Year Capstone

Feb 2026 – Jun 2026

UPTECH · Tunisia - Portal: enterprise after-sales service & product management platform

- Built 4 Spring Boot microservices over MySQL - after-sales service requests, product catalogue, user management and events, exposing 40+ REST endpoints consumed by an Angular front-end.
- Implemented Angular screens across after-sales service and product catalogue workflows end to end: UI, service layer, and the data models behind them.
- Integrated Keycloak for centralised authentication (SSO, OAuth2/OIDC) across all services, and MinIO for attachment object storage.
- Containerised and configured my services for the team's GitLab CI → ArgoCD GitOps pipeline on Kubernetes, working alongside the DevOps engineer who owned it.

Full Stack Developer Intern

Jun 2025 – Jul 2025

Mobtakiron · Tunisia - InnovaLearn: interactive educational platform

<https://github.com/NadineMlayeh/Learning-plateform>

- Built and shipped a live React front-end against a NestJS API and PostgreSQL database, supporting three roles: students taking courses, instructors publishing lessons and exercises, and admins viewing real-time usage.
- Designed the data model and the statistical dashboards behind learning-progress tracking - course completion rates, time spent per lesson, exercise scores, and per-student progress over time.

Full Stack Developer Intern

Jul 2024 – Aug 2024

ElectroHamza · Tunisia – ElectroComfort: inventory & sales management platform

<https://github.com/NadineMlayeh/Home-Appliances-Management-Website>

- Built a home-appliance inventory and sales web application (React front-end, Spring Boot back-end, MySQL) replacing the shop's handwritten paper records.
- Implemented customer management, transaction tracking, and invoicing supporting both instalment and cash payment terms, as specified by the client.

SELECTED PROJECTS

Hakki - Tunisian labour-law document Q&A

2026

Python | Streamlit | NumPy | sentence-transformers | BM25 | Gemini API

<https://github.com/NadineMlayeh/RAG-Hakki>

- Built a retrieval-augmented Q&A system over the Tunisian Labour Code: article-boundary chunking, multilingual embeddings, local vector search, hybrid dense/BM25 retrieval, and grounded answers with article citations.
- Evaluated retrieval quality against a 24-question ground-truth set, improving MRR from 0.424 to 0.630 (+49%) through article-boundary chunking and top-5 retrieval.

LIFE - private life-archive web application2026

NestJS | TypeScript | Prisma | PostgreSQL | React | Three.js (R3F) | Docker

<https://github.com/NadineMlayeh/LIFE>

- Built a full-stack personal archive application: REST API in NestJS over Prisma and PostgreSQL, with bcrypt/JWT authentication and single-use expiring email verification, containerised for local development with Docker and a Mailpit mail-catcher.
- Designed a revocable sharing system using cryptographically random links, where read-only views are filtered *server-side* - private fields are never sent to the client rather than hidden in it.
- Rendered the application’s 3D interface entirely from code with React Three Fiber: primitive geometry plus procedurally generated wood, plaster and fabric textures with Sobel-derived normal maps, no imported models or image assets.

Job Tracker - AI-powered application pipeline2025

Next.js | TypeScript | Supabase | Gemini API | Tailwind

<https://github.com/NadineMlayeh/job-applications-tracker>

- Built a full-stack tracker that pastes a job posting and uses the Gemini API to autofill company, position, link, status (Wishlist / Applied / Interview / Offer / Rejected), application method, and follow-up date, with schema-constrained JSON output reviewable before saving.
- Table and kanban views, unlimited custom fields, CV and cover-letter attachments, Supabase auth and real-time storage.

LEADERSHIP & AWARDS

- **Sponsoring Director, ARSII ISIMM** - secured partner sponsorships and co-organised the IT IMPACT Hackathon and Hackathon ACT for ISIMM.
- Nuit de l’Info - 2nd place, “most creative animated website” challenge (2025); competed 2024 & 2025.

EDUCATION

Engineering Degree in Computer Science - Software Engineering ISIMM, Monastir	2023 – 2026
Integrated Preparatory Cycle ISIMM, Monastir	2021 – 2023
Baccalaureate in Computer Science - High Distinction Lycée Jemmal	2020 – 2021

CERTIFICATIONS & LANGUAGES

Certifications: AWS Academy Cloud Foundations (2025) · AWS Academy Cloud Security Foundations (2025) · Hashgraph Developer Course (2025)

Languages: Arabic (native) · English (fluent, C1) · French (fluent, C1)